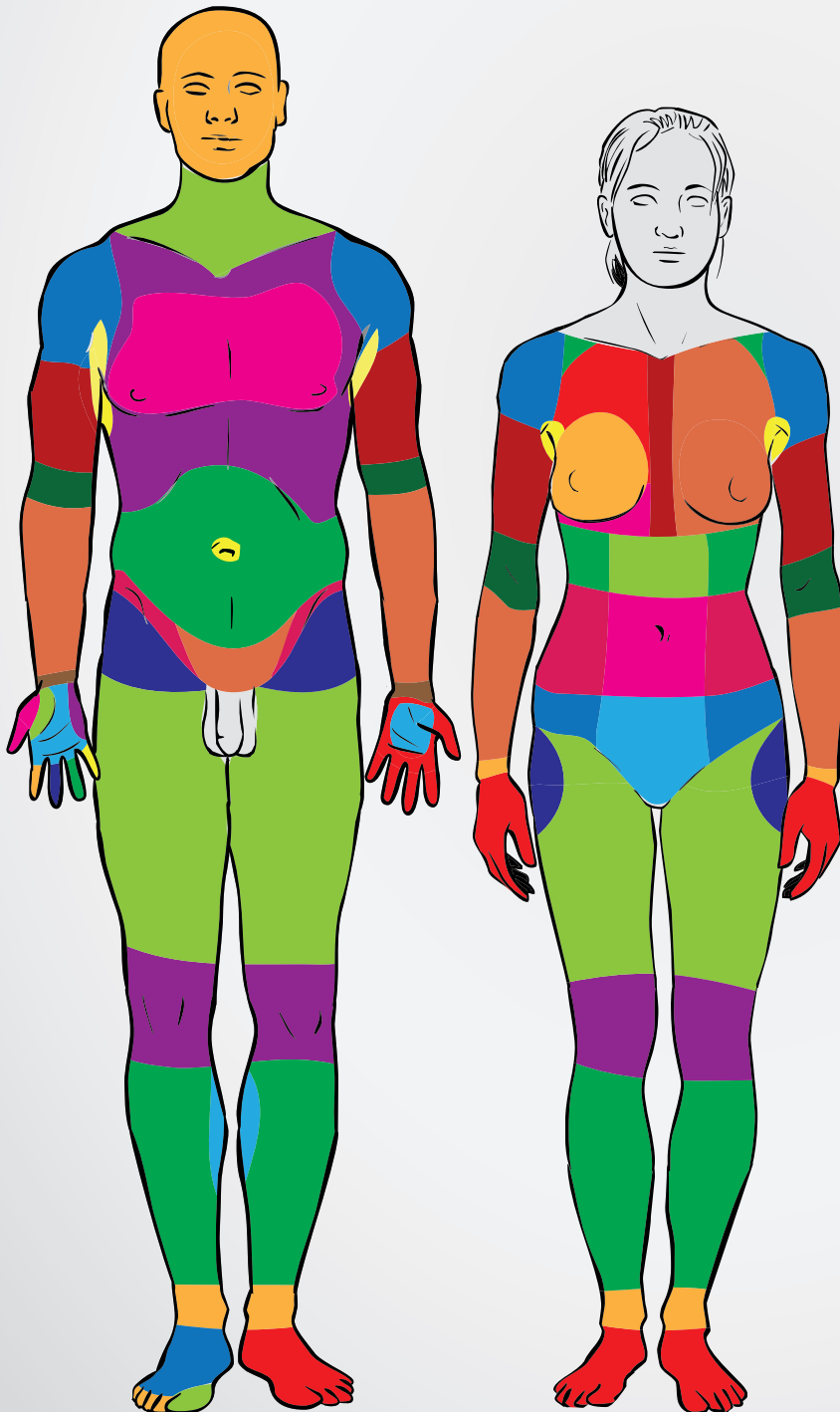


Memorix Anatomy

1

General anatomy



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Anatomy is a scientific field concerned with the structure of the human body. The term “anatomy” comes from the Greek “**anatemnein**”, which literally means “to cut”. Anatomy describes the shape, internal structure and location of the individual parts of the body to enable an understanding of their function, demonstrating that structure and function are inseparably connected. As a **preclinical science**, it provides the foundation for all fields of clinical medicine. We – medical students, physiotherapists, nurses and physicians – would not be able to get by without anatomy and that is why we keep coming back to it.

Definitions

Morphology – a scientific field that includes anatomy, histology and embryology

– a study of the structure and development of the human body

Anatomical terminology – terms describing particular parts of the human body

Anatomical nomenclature – international official anatomical terminology in Latin

Terminologia anatomica (TA 1998) – the most recent version of the anatomical nomenclature

Obsolete terms – terms that have been replaced or removed from the terminologia anatomica

Eponyms – anatomical terms named after living, dead or fictitious people

– terms usually named after the person who first described or discovered the structure

– in clinical medicine some eponyms are used more frequently than the Latin terms

Standard/norm – a set of characteristics occurring most often

Variation – a small deviation from the standard without any functional changes

Anomaly – a significant deviation from the standard causing functional changes

Classifications

According to size:

Macroscopic anatomy – description of the body with the naked eye or with small magnification

Microscopic anatomy/histology – description of the body with the use of microscopic technology

– description of stained slices of tissue viewed under the microscope

According to function:

Normal – study of the structure of the healthy human body

Pathological – study of the anatomical changes that occur with disease

Subspecialities of anatomy:

Systematic anatomy – description of organs according to their development and structure

Topographic anatomy – description of the spatial relationship of different organs to each other and their placement in the body

Others:

Clinical/applied anatomy – use of anatomical findings in clinical medicine

Surface anatomy – description of how the surface of the body appears to the naked eye

– description of body proportions

Comparative anatomy – description of the similarities and differences in the anatomy of humans and animals

World history

1600 B.C.	Egypt	Erbs papyrus the first description of the heart, vessels and some other organs and diseases
460–377 B.C.	Greece	Hippocrates of Kos performed the first human anatomical descriptions
384–322 B.C.	Greece	Aristotle of Stageira described the differences between arteries and veins and performed animal dissections
335–280 B.C.	Greece	Herophilos and his student Erasistratos performed the first human dissection
1 st century B.C.	Greece	Rufus of Ephesos wrote the first book of anatomical terminology
129–205/211	Rome	Galen of Pergamon performed dissections on animals and the first detailed anatomical works
980–1037	Persia	Avicenna wrote “The Canon of Medicine”, the definitive textbook of medicine in the Middle Ages
1213–1288	Egypt	Ibn al-Nafis wrote the first description of the pulmonary circulation and capillaries
1514–1565	Italy	Andreas Vesalius wrote the first modern anatomy textbook “De humani corporis fabrica” in 1543

Anatomy is a descriptive science that almost universally uses mainly **Latin and Greek terms**. Every term has an **English equivalent**. Many of the English equivalents are similar to the Latin, but some are completely different. As Latin is still the first language of medical terminology, **it is important to know at least the basic terms in both languages**.

Basic terms (latin – english)

Abdomen – abdomen, belly	Et – and	Maximus – the biggest	Seu – i.e / in other words
Ala – wing	Excavatio – pouch	Medulla – marrow	Sinister – left
Angulus – angle	Extensor – extensor (muscle)	Membrum – limb	Sinus – cavity, hollow, sinus
Apex – apex, top	Extremitas – extremity	Minimus – the smallest	Septum – enclosed place, septum
Arcus – arc	Facies – surface	Minor – lesser	Spatium – space, room
Arteria – artery	Fissura – fissure	Musculus – muscle	Spina – spine
Articulatio – joint	Flexor – flexor (muscle)	Nervus – nerve	Sulcus – groove
Brevis – short	Foramen – foramen, opening	Nucleus – nucleus	Superficialis – shallow, superficial
Canalis – canal	Fossa – fossa, hole, pit	Nutritium – nourishing	Recessus – space, recess
Capitulum – small head	Fovea – pit, snare	Nodus – node	Tendo – tendon
Caput – head	Ganglion – ganglion	Obliquus – oblique	Teres – rounded, teres
Cartilago – cartilage	Glandula – gland	Os – bone	Thorax – chest, thorax
Cauda – tail	Hallux – toe	Os – mouth	Tonsilla – tonsil
Cavitas – cavity	Hiatus – opening in organ	Ostium – entrance, opening	Transversalis – transverse
Caecus – blind	Impressio – impression	Paries – wall	Truncus – trunk
Cerebrum – brain	Incisura – notch	Pars – part	Tuber – hump, tuber
Cervix – neck	Interosseus – located between bones	Parvus – small	Tuberculum – small bump, tubercle
Circumferentia – circumference	Labium – lip	Pelvis – pelvis	Tuberositas – tuberosity
Collum – neck	Latus – flank	Pectoralis – pectoral, related to chest	Vas, vasa – vessel
Cornu – horn	Ligamentum – ligament	Pectus – front of chest	Vena – vein
Corpus – body	Linea – line	Planum – plane	Venter – belly
Costa – rib	Longitudinalis – longitude	Plexus – plexus	Vesica – bladder
Cranium – cranium	Longus – long	Pollex – thumb	Vertebra – vertebra
Crista – crest	Lympha – lymph	Processus – process	
Dexter – right	Magnus – big	Profundus – deep, profound	
Dorsum – back, dorsum	Major – greater	Prominentia – prominence	
Ductus – duct	Manus – hand	Ramus – branch	
Eminentia – prominence, eminence	Margo – border	Rima – crack, fissure	

Basic abbreviations

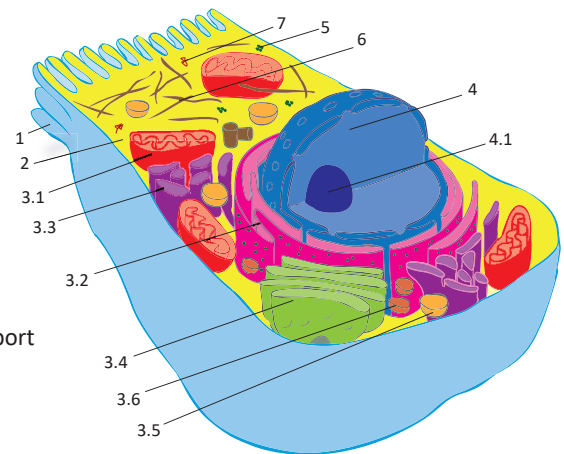
a. – aa. (arteria – arteriae)	ggl. (ganglion)	dx. (dexter)
v. – vv. (vena – venae)	pl. (plexus)	sin. (sinister)
n. – nn. (nervus – nervi)	ant. (anterior)	l. sin. (lateris sinistri) – of the left side
lig. – ligg. (ligamentum – ligamenta)	post. (posterior)	l. dx. (lateris dextri) – of the right side
m. – mm. (musculus – musculi)	int. (internus)	ANS – autonomic nervous system
proc. – procc. (processus – processus)	ext. (externus)	CNS – central nervous system
r. – rr. (ramus – rami)	med. (medialis)	ENS – enteric nervous system
n. l. – nn. l. (nodus lymphoideus – nodi lymphoidei)	lat. (lateralis)	PNS – peripheral nervous system
art. – artt. (articulatio – articulationes)	sup. (superior)	GIT – gastrointestinal tract
ncl. – ncll. (nucleus – nuclei)	inf. (inferior)	UGT – urogenital tract
gl. (glandula)	prof. (profundus)	UL / LL – upper limb / lower limb
	supf. (superficialis)	

Histology is a science that describes cells, tissues, organs and organ systems on a microscopic level. **The cell is the fundamental unit of the body.** During development cells come together to form tissues. **A tissue** is a set of cells and extracellular matrix with a specific function. **There are four types of tissues** that combined together form **organs**. A set of organs forms an **organ system**, which is characterised by its specific functional characteristics.

The cell (*celula*)

The eukaryotic animal cell is the structural and functional unit of the human body.

- **1 Plasmalemma / cell membrane** – the envelope of the cell
 - a biomembrane made up of a phospholipid bilayer
 - is semi-permeable and regulates the transport of substances into and out of the cell
- **2 Cytoplasm** – contains the cytosol, organelles, cytoskeleton and cellular inclusions
- **3 Organelles** – structures in the cytoplasm that are surrounded by biomembranes
 - **3.1 Mitochondria** – produce energy for the cell
 - **3.2 Rough/granular endoplasmic reticulum (RER)** – contains ribosomes on its surface
 - produces and modifies proteins, which are destined to become parts of the cell or are exported into the extracellular matrix
 - **3.3 Smooth/agranular endoplasmic reticulum (SER)** – synthesises steroids, membrane proteins and glycogen
 - the SER in liver hepatocytes carries out detoxification
 - the SER in skeletal muscle cells stores calcium ions
 - **3.4 Golgi complex** – performs posttranslational modification and packaging and secretion of proteins
 - **3.5 Lysosome** – degrades intracellular and extracellular proteins
 - **3.6 Peroxisome** – contains a lot of enzymes and takes part in detoxification
 - degrades fatty acids and amino acids
- **4 Nucleus** – contains genetic material in the form of chromatin
 - **4.1 Nucleolus** – a structure within the nucleus that synthesises ribosomes
- **5 Ribosome** – is located in the cytoplasm and creates proteins for the needs of the cell
- **6 Cytoskeleton** – a network that supports the shape and functions of the cell
 - **6.1 Microfilaments** – orchestrate cellular movement and intracellular transport
 - **6.2 Intermediate filaments** – provide integrity and support to the cell
 - **6.3 Microtubules** – facilitate intracellular transport
- **7 Cytoplasmic inclusions** – deposits of lipids, saccharides, proteins or pigments
 - are either produced by the cell or acquired from outside the cell



Eukaryotic cell

Tissues (*textus*)

- 1 **Epithelial tissue** – an avascular tissue, made of tightly connected cells and very little extracellular material
 - is separated from other tissues by the basal lamina
 - has three major characteristics: number of cell layers, height of the cells and cell surface apparatus (microvilli, kinocilia and stereocilia)
 - 1.1 **Surface epithelium** – has a protecting and supporting function, forms mucous membranes and covers the surface of the body
 - 1.2 **Glandular epithelium** – makes up endocrine glands, which secrete their contents into the blood and exocrine glands, which secrete their contents into body cavities or onto the body surface
 - 1.3 **Specialised epithelium** – includes sensory neuroepithelium and the epithelial part of the thymus
 - 2 **Connective tissue** – has mechanical, protective and supporting functions
 - consists of cells in a variable amount of extracellular material known as the extracellular matrix
 - the extracellular matrix has a fibrillar component of collagen and elastic fibers and an amorphous component of glycosaminoglycans
 - 2.1 **Connective tissue proper** – possesses many cells in an extracellular matrix and has a gel-like consistency
 - 2.1.1 **Loose connective tissue** – fills spaces between organs and can be found in the walls of organs beneath the mucosa
 - 2.1.2 **Dense connective tissue** – makes up ligaments, tendons and articular capsules
 - 2.2 **Cartilage tissue** – is composed of cells in an extracellular matrix and has a solid but elastic consistency
 - 2.2.1 **Hyaline cartilage** – is located on the contact surfaces of bones inside joints and forms epiphyseal growth plates
 - is integral in the process of bone healing, is found in the wall of the trachea and bronchi and forms part of the ribs
 - 2.2.2 **Fibrous cartilage** – is found in intervertebral discs and in the pubic symphysis
 - 2.2.3 **Elastic cartilage** – is found in the auricle of the ear and in the epiglottis
 - 2.3 **Bone/osseous tissue** – is composed of a small proportion of cells in a firm extracellular matrix that is mineralised by calcium
 - 3 **Muscle tissue** – is made up of cells that contain myofibrils (actin and myosin), which give them the ability to contract
 - 3.1 **Smooth muscle tissue** – is found in the walls of organs and vessels and is innervated by the autonomic nervous system
 - 3.2 **Skeletal striated muscle tissue** – is located on the skeleton and innervated by somatomotor cranial and spinal nerves
 - 3.3 **Striated cardiac muscle** – is found only in the heart and has its own electrical conduction system
 - 4 **Nerve tissue** – is made up of neurons and supporting cells (neuroglia)
 - 4.1 **Neurons** – cells that are able to accept, process and transmit impulses
 - 4.2 **Neuroglia** – protect and support neurons
- + **“Liquid/trophic tissue”** – comprises interstitial fluid, blood and lymph

Embryology is a scientific field concerned with the **creation and development of the embryo and foetus, all the way from a fertilised oocyte to birth**. Embryology also deals with anomalies that occur during development that give rise to **congenital defects**. The **3 germ layers** arise during the third week of the development in the form of the **trilaminar germ disc**, which comprises the endoderm, mesoderm and ectoderm. **The pharyngeal arches** originate from mesenchyme as pairs of arc-shaped prominences **in the region of head and neck** of the embryo during the fourth and fifth week of development.

Developmental periods

1. **Prenatal/gestation/intrauterine period** – extends from fertilisation to birth
 - 1.1 **Fertilisation** – the process of fusion of an egg and a sperm to form a zygote
 - 1.2 **Blastogenesis** – occurs during the first two weeks after the fertilisation
 - under unfavourable external influences only two things can happen following the “all or nothing principle”: either the embryo will survive or it will be aborted
 - 1.3 **Embryonic period** – extends from fertilisation to the end of the eighth week
 - all organ systems are formed
 - the embryo is most vulnerable to toxic, radioactive, metabolic, pharmacological and other influences that may cause developmental defects
 - 1.4 **Foetal period** – extends from the end of the 8th week to birth
 - the organ systems continue to grow and develop
2. **Perinatal period** – starts at the beginning of the 28th week of intrauterine development and ends one week after birth
3. **Postnatal period** – extends from birth to death

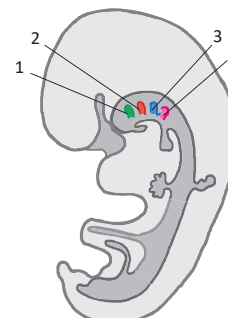
Primary germ layers

- 1 **Ectoderm** – the outer layer of the trilaminar germ disc
 - differentiates to form epithelial tissues including the epithelium of the skin and its appendages (hair, nails and sweat glands), the epithelium of the initial and terminal parts of the gastrointestinal tract, the adenohypophysis and the neurogenic placodes
 - 1.1 **Neuroectoderm** – a derivative of the ectoderm that gives rise to the neural tube and the neural crest
 - the neural tube differentiates into the brain and the spinal cord
 - the cells of the neural crest migrate to various part of the body
 - the neural crest forms the peripheral nerves and ganglia, fibrous tissue in the head and neck, melanocytes, the receptors cells of the carotid body, the medulla of the suprarenal glands and some structures in the heart
- 2 **Mesoderm** – the middle layer of the trilaminar germ disc
 - 2.1 **Paraxial mesoderm** – forms somites, which develop into bones, muscles and the dermis
 - 2.2 **Intermediate mesoderm** – forms the urogenital tract
 - 2.3 **Lateral plate mesoderm** – forms the body wall
 - + **Mesenchyme** – embryonic connective tissue that forms vessels, smooth muscles, haematopoietic tissue and mature connective tissue
- 3 **Endoderm** – the inner layer of the trilaminar germ disc
 - gives rise to the epithelium of the liver, pancreas, gastrointestinal tract (apart from the mouth, nasopharynx and anus), auditory tube, trachea, bronchi, alveoli, urinary bladder, upper part of the urethra, thyroid gland, parathyroid glands and thymus

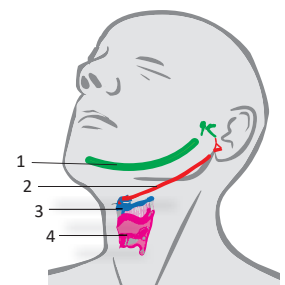
Pharyngeal arches (*arcus pharyngei*)

Each pharyngeal arch contains an artery, muscle tissue, a cranial nerve and a connective tissue element of cartilage, ligament or bone.

- 1 **First pharyngeal arch**
 - bones:** mandible, maxilla, zygomatic bone, malleus and incus
 - muscles:** masticatory muscles, anterior belly of the digastricus, mylohyoid, tensor veli palatini and tensor tympani
 - cranial nerve:** trigeminal nerve
 - artery:** terminal part of the maxillary artery
- 2 **Second pharyngeal arch**
 - bones:** stapes, styloid process and the lesser horn and upper part of the body of the hyoid bone
 - muscles:** mimetic muscles, platysma, stapedius, stylohyoid and the posterior belly of the digastricus
 - cranial nerve:** facial nerve; **artery:** stapedia artery
- 3 **Third pharyngeal arch**
 - bones:** greater horn and the lower part of the body of the hyoid bone
 - muscles:** stylopharyngeus
 - cranial nerve:** glossopharyngeal nerve;
 - artery:** common carotid artery and the first part of the internal carotid artery
- 4 **Fourth and sixth pharyngeal arch**
 - cartilage:** laryngeal skeleton
 - muscles:** musculature of the pharynx and larynx
 - cranial nerve:** vagus nerve - the superior laryngeal nerve is from the 4th arch and the recurrent laryngeal nerve is from the 6th arch
 - artery:** the aortic arch and right subclavian artery are from the 4th arch and the pulmonary arteries and ductus arteriosus are from the 6th arch



Pharyngeal arches in the embryo



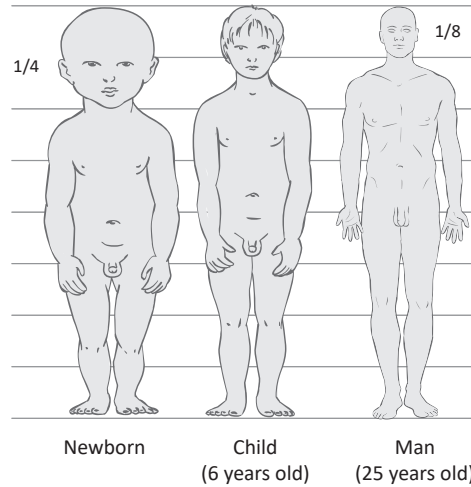
Derivatives of the pharyngeal arches

In comparison to other animals, **the human undergoes an extraordinary process of growth and development during childhood and puberty.** The two major periods of acceleration of growth are in **infancy** and **puberty**. Growth in-between these periods is slower. This allows time for the child to develop psychologically before undergoing **full physical transformation into adulthood.** Organs and organ systems grow both **isometrically** (proportionally) and **allometrically** (disproportionally) in relation to other organs and organ systems.

Head

Proportions of the head to the rest of the body

- the head forms half the length of the embryo in the 6th–7th week of intrauterine development
- the length of the head equals one quarter the length of the newborn at birth
- the circumference of the head and thorax are equal by the 5th month of life and from this moment onwards the thorax will grow faster
- the head is the fastest growing part of the body in the 1st year of life
- the head forms about one eighth the length of the body in adulthood



Thorax

Heart – in childhood, the proportion of the weight of the heart to the rest of the body is about twice as high as it is in adulthood

Lungs and respiratory passage – the lower respiratory tract elongates until the 1st year and then enlarges only in diameter
– the number of alveoli increases until the 8th year of life and then only increase in size

Oesophagus – is about half the length of the trunk in newborns
– is about one quarter the length of the trunk in adults

Abdomen

Stomach – the ratio of the weight of the stomach to the rest of the body is highest in children

Liver – forms about 5 % of body weight in children and about 2.5 % of body weight in adults

Kidneys – the final number of nephrons is reached by the 35th–36th week of intrauterine development

Testes – are formed in the lumbar region
– in the 3rd month of intrauterine development, the unequal growth of the thorax and the traction of the gubernaculum testis begin to move the testes caudally through the retroperitoneum and inguinal canal into the scrotum
– by the 8th month of intrauterine development, the testes are fully descended in the scrotum

Central nervous system

The weight of the brain – about 300–400 g in newborns or approximately 10 % of total body weight

- gyri and sulci appear on the surface of the cerebral cortex in the 4th month of intrauterine life
- reaches 75 % of the weight of an adult brain by the end of the 1st year of life
- reaches 100 % the weight of an adult brain in the 15th year of life
- the adult brain weighs 1300–1400 g, which is approximately 2% of total body weight
- the increase in brain weight after birth is caused mainly by myelination and an increase in the number of neuroglial cells

Spinal cord – is as long as the vertebral canal at the start of embryonic development

- the vertebral canal grows faster than the spinal cord from the 3rd month of intrauterine development
- the spinal cord therefore does not fill out the whole length of the vertebral canal
- the final level of the caudal border of the spinal cord is L1 – L2

The anterior fontanelle (*fonticulus anterior*) closes during the 1st–3rd year of life. **The posterior fontanelle** (*fonticulus posterior*) usually closes by the 3rd month of life. However it may already be closed at birth.

The volume of cerebrospinal fluid in a newborn born by a spontaneous vaginal delivery is about half that of a newborn born by a Caesarean section. During the baby's passage through the birth canal, the cerebrospinal fluid is forced into the venous sinuses.

The menarche is the onset of menstruation in girls. It is one of the first signs of puberty and appears between the 10th–17th year of age. In Central Europe, it usually occurs between the 12th–13th year of age. In the last couple of decades sexual maturation has been occurring progressively earlier.

The onset of puberty has been occurring earlier in life and the average height of the population has been steadily increasing over the last century. This phenomenon, known as "**the secular trend of puberty**" has been attributed to a better quality of nutrition.

The skull of a newborn is discussed on page 35.

The descent of testes is discussed on page 241.

The anatomy of pregnancy, the embryo and birth are discussed on page 257.

Vertebromedullary topography is described on page 398.

Psychomotor development is discussed on page 478.

The breast and mammary gland is described on page 517.

Clinical notes

A prominent pulsating fontanelle can be a sign of increased intracranial pressure, meningitis or other serious conditions. A sunken fontanelle on the other hand can indicate poor hydration of the baby.

Premature puberty (*pubertas praecox*) is characterised by onset of secondary sex characteristics before the 8th year in girls and before the 9th year in boys. It occurs in about 0.6 % of all children, more often in girls.

Pubertas tarda is a term for delayed puberty.

During puberty, the process of **psychosomatic maturation** of a child into an adult takes place. This process is accompanied by a set of **anatomical and physiological changes** that are integrally linked to each other. Using these changes to evaluate the stage of puberty is far more informative than relying solely on the age of the child. The onset of these changes can vary in children of the same sex in the range of 3–4 years. Generally speaking, **puberty starts in males 1–2.5 years later than in females**. The **Tanner scale** is commonly used to assess the level of changes occurring during puberty. In males, the growth of pubic hair, testes, penis and scrotum are evaluated. In females, development of breasts and the growth of pubic hair are evaluated.

Pubic hair in both genders

Stage Ph1 (< 10 years of age)

- no pubic hair

Stage Ph 2 (10–11.5 years of age)

- a small amounts of long fine weakly pigmented hair located around the root of the penis and the scrotum in males and on the mons pubis and labia majora in females

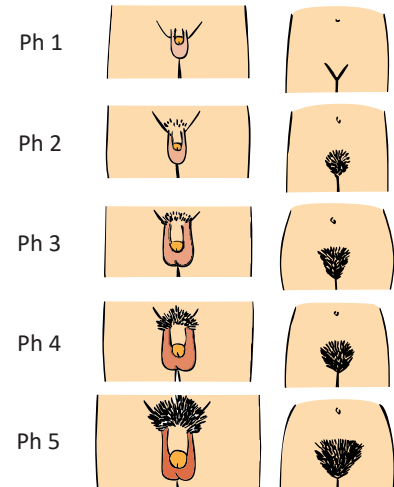
Stage Ph 3 (11.5–13 years of age) – the hair becomes darker, thicker and curly and spreads out on the skin overlying the pubic symphysis

Stage Ph 4 (13–15 years of age)

- the pubic hair is of adult type in quality, but in a smaller amount than in an adult
- in girls, the pubic hair covers a triangular shaped area of skin on the mons pubis and labia majora

Stage Ph 5 (>15 years of age)

- pubic hair of an adult, grows on adjacent parts of thighs
- female type: the upper border of the pubic hair is sharp and horizontally straight
- male type: usually continues cranially in a narrow triangle reaching as high as the navel



Male external genital organs

Stage G 1 (< 9 years of age)

- the volume of the testes is less than 1.5 ml
- the size of penis is 3 cm or less

Stage G 2 (9–11 years of age) – the volume of the testes is 1.6–6 ml

- the scrotum and testes enlarge
- the skin of the scrotum becomes thinner, wider and reddish
- the length of the penis remains unchanged

Stage G 3 (11–12.5 years of age) – the volume of the testes is 6–12 ml

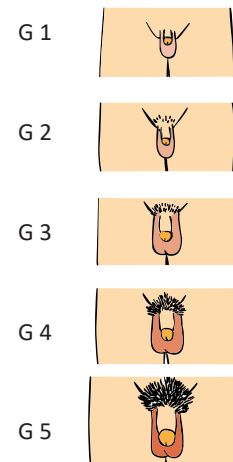
- the scrotum and penis enlarge
- the penis grows to about 6 cm in length

Stage G 4 (12.5–14 years of age) – the volume of the testes is 12–20 ml

- the scrotum enlarges and becomes darker coloured
- the penis grows to about 10 cm in length
- the rate of growth of the width of the penis increases

Stage G 5 (> 14 years of age) – the volume of the testes is greater than 20 ml

- the external genital organs reach adult size and shape
- the length of the penis is approximately 15 cm



Male external genital organs

Female breast

Stage M 1 (<10 years of age) – *mamma puerilis* / *mamma neutralis*

- no gland tissue is present
- the unpigmented areola is in the same plane as the skin

Stage M 2 (10–11.5 years of age) – *mamma areolata*

- a small mass of glandular tissue is palpable in the area of the areola

Stage M 3 (11.5–13 years of age) – the breast is pointy and cone-shaped

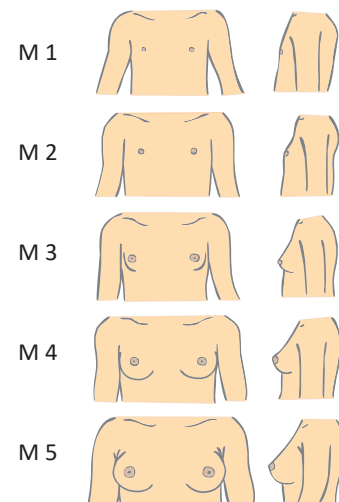
- the gland grows beyond the margins of the areola

Stage M 4 (13–15 years of age) – the areola is raised above the plane of the skin

- the areola is slightly pigmented
- the areolar tubercles of the areolar glands are not visible
- the nipple is not yet erectile

Stage M 5 (>15 years of age) – the definitive appearance of a mature breast

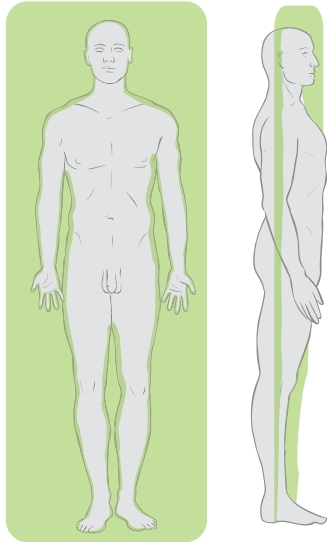
- the areola is again in the same plane as the skin
- the areola is hyperpigmented
- the areolar tubercles are visible on the surface of the areola
- the nipple is erectile



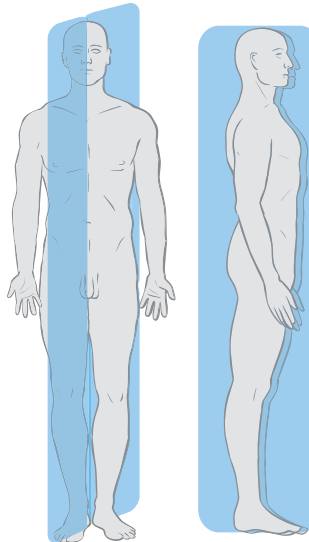
Female breast

To avoid confusion, the structures of the human body are described in a specific posture. This posture is called **the anatomical position** and is an **upright posture with the upper limbs on the sides of the body with the palms facing forward**. To help us orientate structures on the human body, we use **three principle perpendicular planes of reference**. These are the sagittal plane, the frontal plane and the transverse plane. **The sagittal plane** runs through the body from the back to the front. **The frontal plane** is parallel to the forehead and divides the body into posterior and anterior portions. **The transverse plane** runs through the body horizontally, dividing it into cranial and caudal portions. **The mid-sagittal or median plane** is a sagittal plane found **exactly at the midpoint of the body** and cuts the body into **symmetrical left and right halves**.

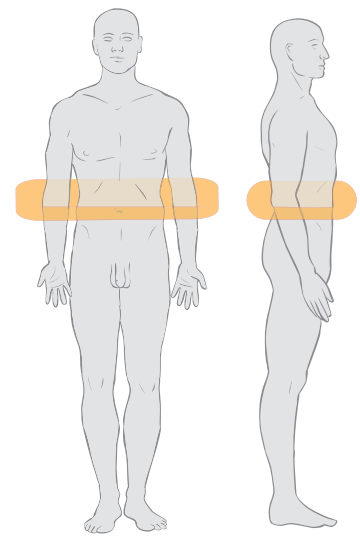
Planes of the human body (*plana corporis humani*)



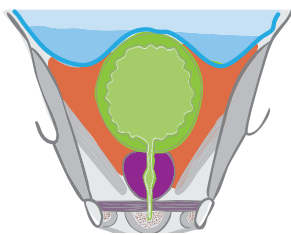
● **Frontal plane / coronal plane**
(*planum frontale*)



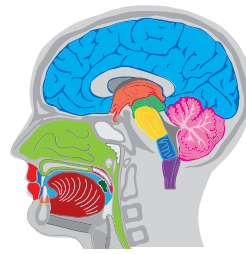
● **Sagittal plane** (*planum sagittale*)



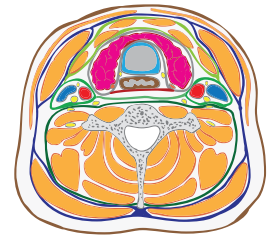
● **Transverse plane / horizontal plane / axial plane / cross-section** (*planum transversarium*)



Frontal section of the male pelvis



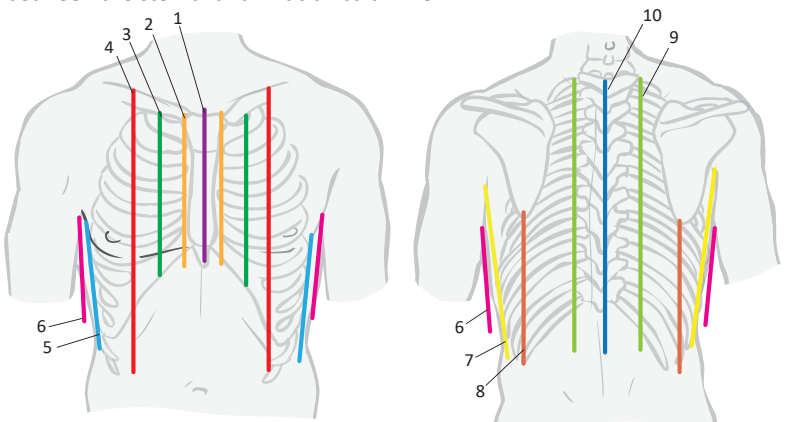
Sagittal section of the head



Transverse section of the neck

Lines of the human body (*lineae corporis humani*)

- 1 **Anterior median line** (*linea mediana anterior*) – runs vertically through the centre of the thorax and abdomen
- 2 **Sternal line** (*linea sternalis*) – runs vertically along the lateral side of the sternum
- 3 **Parasternal line** (*linea parasternalis*) – runs vertically in-between the sternal and midclavicular line
- 4 **Midclavicular line** (*linea medioclavicularis*) – runs vertically through the midpoint of the clavicle
- 5 **Anterior axillary line** (*linea axillaris anterior*) – runs vertically through the anterior axillary skin fold
- 6 **Midaxillary line** (*linea axillaris media*) – runs vertically through the apex of the axilla
- 7 **Posterior axillary line** (*linea axillaris posterior*) – runs vertically through the posterior axillary skin fold
- 8 **Scapular line** (*linea scapularis*) – runs vertically through the middle of the inferior angle of the scapula
- 9 **Paravertebral line** (*linea paravertebralis*) – runs vertically along the side of the vertebral column
- 10 **Posterior median line** (*linea mediana posterior*) – runs vertically through the middle of the vertebral column, passing through the spinous processes of the vertebrae

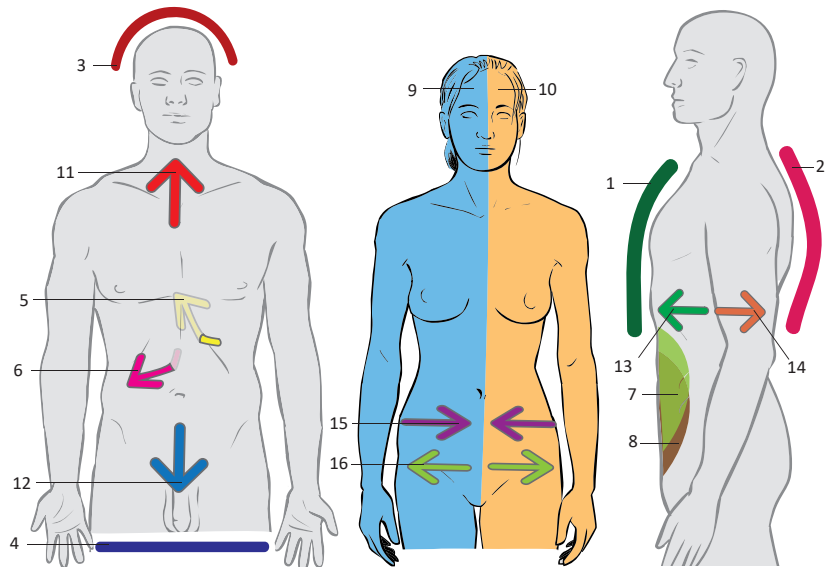


Lines of the human body

Locations and directions are basic orientation descriptions used in both gross and clinical anatomy. They are not only used to describe individual organs but are also used in **topographical anatomy** to describe the spatial relationships of one organ to another.

Locations and directions used on the whole body

- 1 **Anterior** – towards the front
- 2 **Posterior** – towards the back
- 3 **Superior** – upwards
- 4 **Inferior** – downwards
- 5 **Internal/inner** (*internus*)
- 6 **External/outer** (*externus*)
- 7 **Superficial** (*superficialis*)
- 8 **Deep** (*profundus*)
- 9 **Right** (*dexter*)
- 10 **Left** (*sinister*)
- 11 **Cranial** (*cranialis*) – towards the head
- 12 **Caudal** (*caudalis*) – towards the tail bone
- 13 **Ventral** (*ventralis*) – towards the front
- 14 **Dorsal** (*dorsalis*) – towards the back
- 15 **Medial** (*medialis*) – towards the centre of the body or towards the axis of the limb
- 16 **Lateral** (*lateralis*) – away from the centre of the body or away from the axis of the limb
- 17 **Ipsilateral** – refers to something on the same side of the body
- 18 **Contralateral** – refers to something on the opposite side of the body



Locations and directions used on the limbs

Directions on the upper and lower limbs:

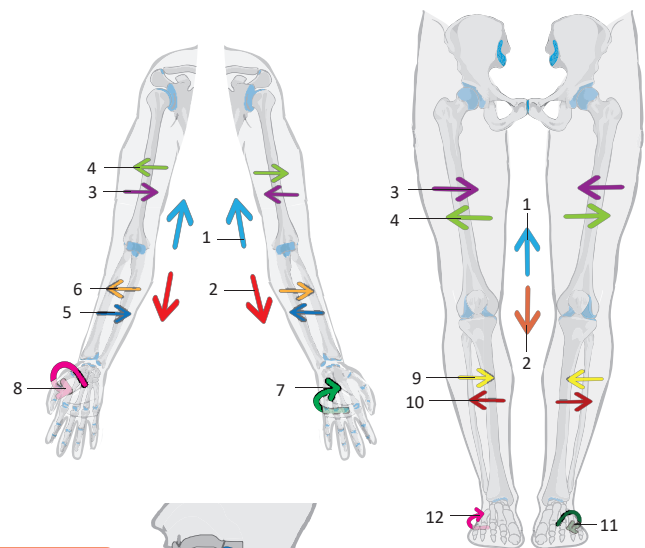
- 1 **Proximal** (*proximalis*) – towards the trunk
- 2 **Distal** (*distalis*) – away from the trunk
- 3 **Medial** (*medialis*) – towards the middle axis
- 4 **Lateral** (*lateralis*) – away from the middle axis

Directions on the upper limb:

- 5 **Ulnar** (*ulnaris*) – on or towards the ulnar side of the forearm
- 6 **Radial** (*radialis*) – on or towards the radial side of the forearm
- 7 **Palmar/volar** (*palmaris/volaris*) – on or towards the palm
- 8 **Dorsal** (*dorsalis*) – on or towards the dorsum of the hand

Directions on the lower limb:

- 9 **Tibial** (*tibialis*) – on or towards the tibial side of the leg
- 10 **Fibular/peroneal** (*fibularis/peroneus*) – on or towards the fibular side of the leg
- 11 **Plantar** (*plantaris*) – on or towards the sole
- 12 **Dorsal** (*dorsalis*) – on or towards the dorsum of the foot



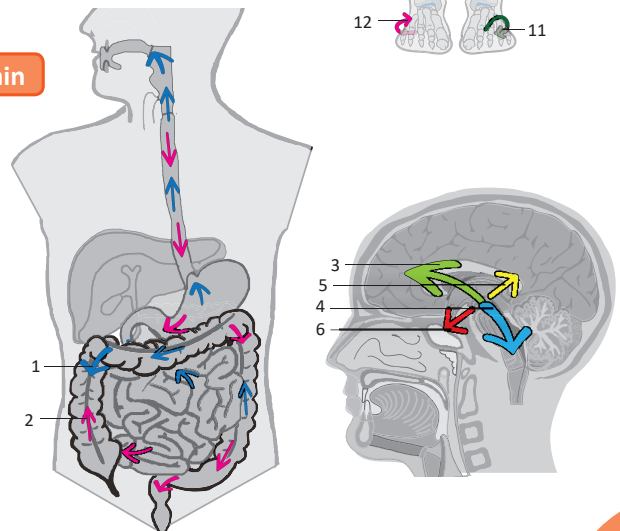
Terms of direction specific for the gastrointestinal tract and the brain

Directions in the gastrointestinal tract:

- 1 **Oral** (*oralis*) – towards the oral cavity
- 2 **Aboral** (*aboralis*) – towards the anal canal

Directions in the brain:

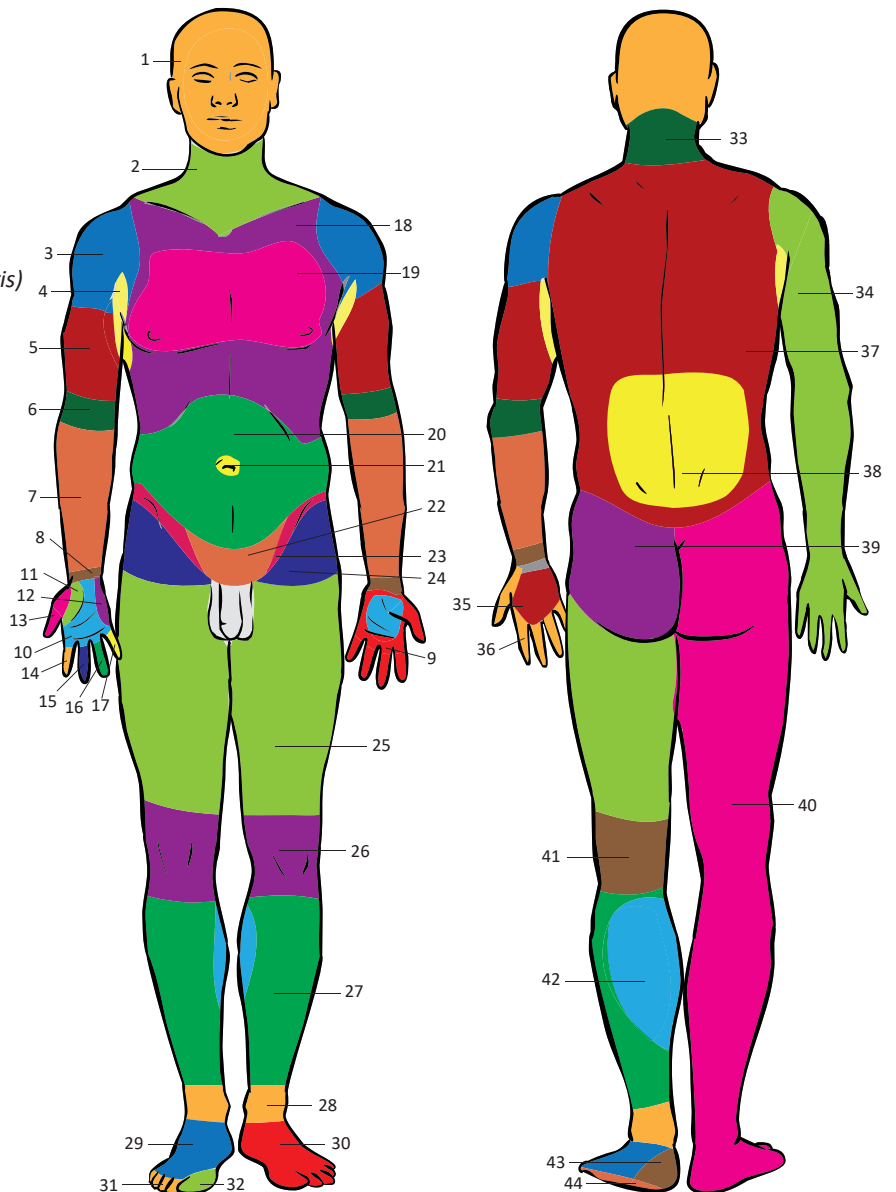
- 3 **Rostral/frontal/cranial** (*rostralis/frontalis/cranialis*) – towards the forehead
- 4 **Caudal/occipital** (*caudalis/occipitalis*) – towards the occiput
- 5 **Dorsal** (*dorsalis*) – towards the cranial vault (calvaria)
- 6 **Ventral/basal** (*ventralis/basalis*) – towards the cranial base



The following terms are used to describe **specific parts of the human body in the standard anatomical position**. **Constitutional typology** describes body shape according to the height and width of the body, length of the limbs and the amount of muscle and body fat.

Parts of the human body (*partes corporis humani*)

- 1 Head (*caput*)
- 2 Neck (*collum*)
- 3 Shoulder (*omos*)
- 4 Axilla, armpit (*axilla*)
- 5 Arm (*brachium*)
- 6 Elbow (*cubitus*)
- 7 Forearm (*antebrachium*)
- 8 Wrist (*carpus*)
- 9 Hand (*manus*)
- 10 Palm (*palma, vola*)
- 11 Thenar eminence (*thenar / eminentia thenaris*)
- 12 Hypothenar eminence (*hypothenar / eminentia hypothenaris*)
- 13 Thumb (*pollex / digitus primus*)
- 14 Index finger (*index / digitus secundus*)
- 15 Middle finger (*digitus medius/tertius*)
- 16 Ring finger (*digitus anularis/quartus*)
- 17 Little finger (*digitus minimus/quintus*)
- 18 Thorax
- 19 Front of chest (*pectus*)
- 20 Abdomen
- 21 Navel (*umbilicus*)
- 22 Pelvis
- 23 Groin (*inguen*)
- 24 Hip (*coxa*)
- 25 Thigh (*femur*)
- 26 Knee (*genu*)
- 27 Leg (*crus*)
- 28 Ankle (*tarsus*)
- 29 Metatarsus
- 30 Foot (*pes*)
- 31 Toes (*digiti pedis*)
- 32 Big toe (*hallux / digitus primus*)
- 33 Nape of neck (*nucha*)
- 34 Upper limb (*membrum superius*)
- 35 Dorsum of hand (*dorsum manus*)
- 36 Fingers (*digiti manus*)
- 37 Back (*dorsum*)
- 38 Loin (*lumbus*)
- 39 Buttocks (*nates/clunes*)
- 40 Lower limb (*membrum inferius*)
- 41 Posterior side of knee (*poples*)
- 42 Calf (*sura*)
- 43 Heel (*calx*)
- 44 Sole (*planta*)



Constitutional typology

Asthenic type / ectomorph

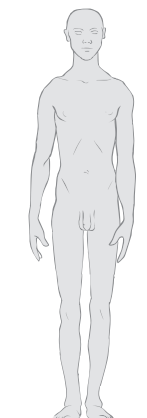
- weak skeleton, poorly developed muscles
- slim trunk, long limbs, small head and oblong shaped face

Athletic type / mesomorph

- strong skeleton, well developed muscles
- long head with flat nose
- prominent bones of the face forming sharp facial features

Pyknic type / endomorph

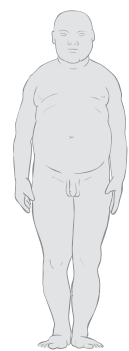
- fat stocky figure
- round head with a wide straight nose



Asthenic type



Athletic type



Pyknic type

Regions of the human body (*regiones corporis humani*)

Regions of the head (*regiones capitis*):

- 1 Frontal region (*regio frontalis*)
- 2 Parietal region (*regio parietalis*)
- 3 Occipital region (*regio occipitalis*)
- 4 Temporal region (*regio temporalis*)
- 5 Auricular region (*regio auricularis*)
- 6 Mastoid region (*Regio mastoidea*)

Facial regions (*regio facialis*):

- 7 Orbital region (*regio orbitalis*)
- 8 Infra-orbital region (*regio infraorbitalis*)
- 9 Buccal region (*regio buccalis*)
- 10 Parotid region (*regio parotideomasseterica*)
- 11 Zygomatic region (*regio zygomatica*)
- 12 Nasal region (*regio nasalis*)
- 13 Oral region (*regio oralis*)
- 14 Mental region (*regio mentalis*)

Regions of the neck (*regiones cervicales*):

- 15 Anterior cervical region/triangle (*regio cervicalis anterior / trigonum cervicale anterius*)
 - 15.1 Submental triangle (*trigonum submentale*)
 - 15.2 Submandibular triangle (*trigonum submandibulare*)
 - 15.3 Carotid triangle (*trigonum caroticum*)
 - 15.4 Muscular/omotracheal triangle (*trigonum musculare/omotracheale*)
- 16 Sternocleidomastoid region (*regio sternocleidomastoidea*)
 - 16.1 Lesser supraclavicular fossa (*fossa supraclavicularis minor*)
- 17 Lateral cervical region / posterior triangle of the neck (*regio cervicalis lateralis / trigonum cervicale laterale*)
 - 17.1 Omoclavicular triangle / subclavian triangle (*trigonum omoclaviculare*)
 - 17.2 Omotracheoid triangle (*trigonum omotracheoidium*)
- 18 Posterior cervical region (*regio cervicalis posterior*)

Thoracic regions (*regiones thoracicae*):

- 1 Parasternal region (*regio presternalis*)
- 2 Clavicopectoral/deltopectoral triangle (*trigonum clavipectorale/deltopectorale*)
- 3 Pectoral region (*regio pectoralis*)
- 4 Mammary region (*regio mammaria*)
- 5 Inframammary region (*regio inframammaria*)
- 6 Axillary region (*regio axillaris*)

Abdominal regions (*regiones abdominales*):

- 7 Epigastric region (*regio epigastrica / epigastrium*)
- 8 Hypochondrium (*regio hypochondriaca / hypochondrium*)
- 9 Umbilical region (*regio umbilicalis / mesogastrium*)
- 10 Lateral region (*regio lateralis / mesogastrium*)
- 11 Pubic region (*regio pubica / hypogastrium*)
- 12 Inguinal region (*regio inguinalis / hypogastrium*)

Regions of the back (*regiones dorsales*):

- 13 Vertebral region (*regio vertebralis*)
- 14 Scapular region (*regio scapularis*)
- 15 Infrascapular region (*regio infrascapularis*)
- 16 Lumbar region (*regio lumbalis*)
- 17 Sacral region (*regio sacralis*)

Perineal region (*regio perinealis*):

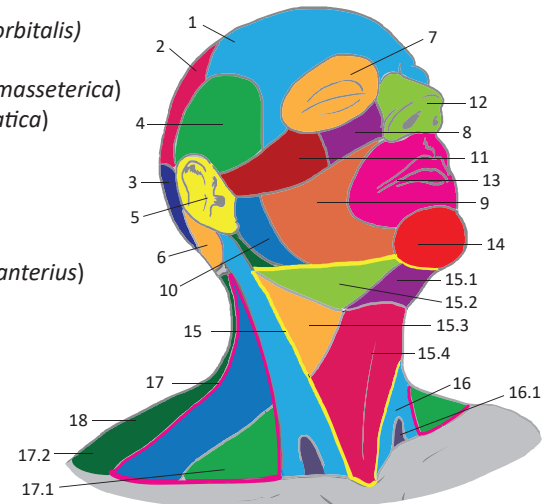
- 18 Anal triangle (*regio analis*)
- 19 Urogenital triangle (*regio urogenitalis*)

Regions of the upper limb (*regiones membri superioris*):

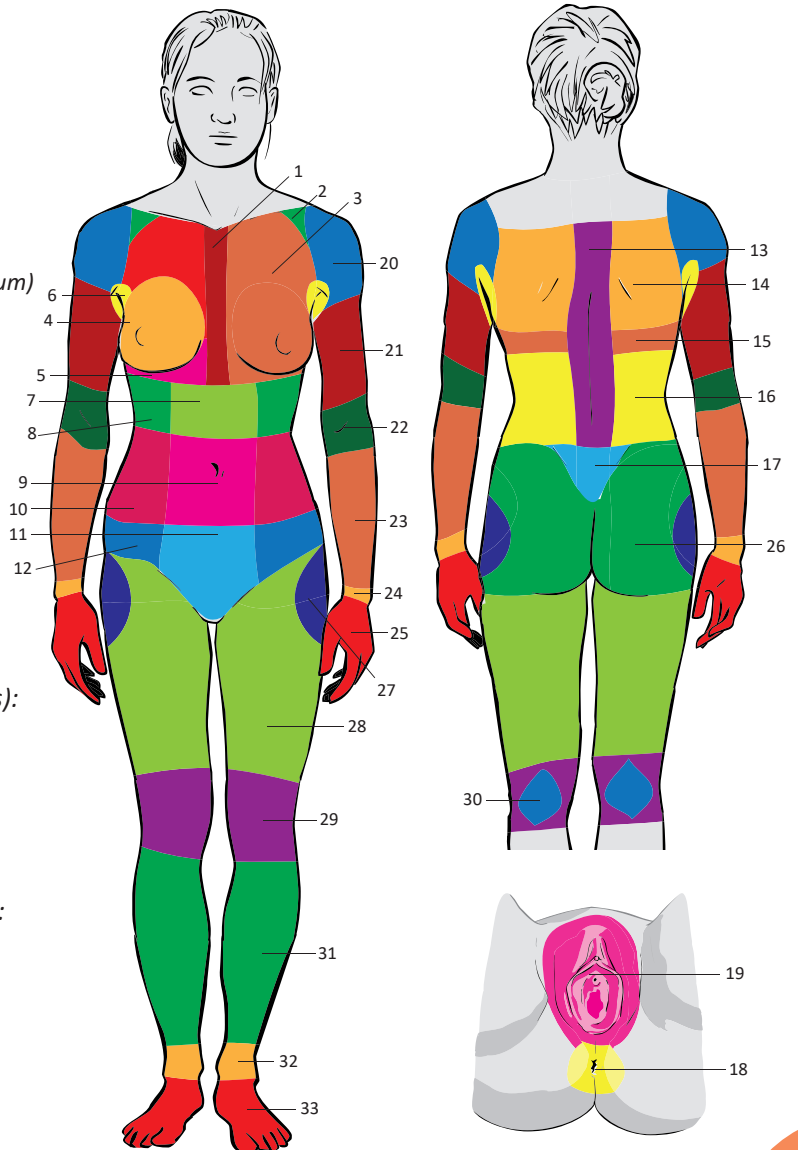
- 20 Deltoid region (*regio deltoidea*)
- 21 Brachial region (*regio brachialis*)
- 22 Cubital region (*regio cubitalis*)
- 23 Antebrachial region (*regio antebrachialis*)
- 24 Carpal region (*regio carpalis*)
- 25 Hand region (*regio manus*)

Regions of the lower limb (*regiones membri inferioris*):

- 26 Gluteal region (*regio glutealis*)
- 27 Hip region (*regio coxae*)
- 28 Femoral region (*regio femoris*)
- 29 Knee region (*regio genus*)
- 30 Popliteal region (*regio poplitea*)
- 31 Leg region (*regio cruris*)
- 32 Ankle region (*regio tarsalis*)
- 33 Foot region (*regio pedis*)



Regions of the head and neck



Anatomical terminology uses **eponyms**. Eponyms are terms named after the person that discovered or described the anatomical structure. Their use lies **predominantly in clinical anatomy**.

1. **ACHILLES** **Achilles' tendon** – the tendon of the triceps surae, named after a Greek mythological warrior
2. **ALCOCK** **Alcock's canal** – a space on the inner side of the obturator internus
3. **ARANTIUS** **Arantius' duct** – a shunt found in the foetus that allows blood to pass from the umbilical vein to the inferior vena cava, bypassing the liver
4. **ASCHOFF** **Aschoff-Tawara's node** – the AV node of the conducting system of the heart
5. **AUERBACH** **Auerbach's myenteric plexus** – a nervous plexus within the muscular layer of the gastrointestinal tract
6. **BARTHOLINUS** **Bartholin's glands** – mucous glands in the vestibule of the vagina
Bartholin's duct – the largest excretory duct of the sublingual salivary gland
7. **BAUHIN** **Bauhin's valve** – a narrowing at the transition between the small and large intestines
8. **BILLROTH** **Cords of Billroth** – the red pulp cords between sinusoids in the spleen
9. **BOCHDALEK** **Bochdalek's triangle** – a triangular defect between the lumbar and costal parts of the diaphragm
Bochdalek' flower spray – a protrusion of the choroid plexus of the fourth ventricle through the lateral aperture into the subarachnoid space
10. **BÖHME** **Cannon-Böhm point** – a point located between the middle and left thirds of the transverse colon that marks the transition of parasympathetic innervation from the vagus nerve to the sacral splanchnic nerves
11. **BOTALLO** **Ductus Botalli** – a foetal shunt connecting the pulmonary trunk to the aortic arch
12. **BOWMAN** **Bowman's capsule** – a sac, enclosing the glomerulus, forming the first part of the nephron
13. **BROCA** **Broca's area** – an area of cortex located in the posterior part of the inferior frontal gyrus in the speech dominant hemisphere (Brodman areas 44 and 45)
14. **BRODMANN** **Brodman's map** – a cytoarchitectural map of the cerebral cortex
15. **CAJAL** **Cajal's nucleus** – a group of neurons located in the periaqueductal grey substance of the tegmentum of the midbrain
16. **CALOT** **Calot's triangle** – an anatomical space containing a potential ligation site of the cystic artery
17. **CAMPER** **Fascia of Camper** – a fibrous-adipose layer of the anterior abdominal wall
18. **CHOPART** **Chopart's joint** – a joint line located transversely through the tarsal bones
19. **CLARKE** **Stilling-Clarke's nucleus** – a nucleus located in the posterior horn of the spinal cord
20. **CLAUDIUS** **Claudius' fossa** – a slight depression that houses the ovary, located between the internal iliac vessels and the sacrum
21. **CLOQUET** **Cloquet's node** – the most proximal of the deep inguinal lymph nodes, located in the lacuna vasorum
22. **COCKETT** **Cockett's perforators** – three connections between the superficial and deep venous system, located on the medial side of the calf
23. **CORTI** **Organ of Corti** – the auditory organ proper
24. **DOUGLAS** **Pouch of Douglas** – the deepest space of the peritoneal cavity in women
25. **EUSTACHIO** **Eustachian tube** – the auditory tube, a connection between the nasopharynx and tympanic cavity,
Eustachian valve – the valve of the inferior vena cava, located in the right atrium of the heart
26. **FALLOPPIO** **Fallopian canal** – the part of the facial canal located in the petrous part of the temporal bone
Fallopian tube – the uterine tube
27. **FLACK** **Keith-Flack's node** – the SA node of the conducting system of the heart
28. **FROHSE** **Frohse's arcade** – a fibrous arch in the supinator canal
29. **GALENOS** **Vein of Galen** – the largest cerebral vein, drains blood from the deep parts of the brain into the straight sinus
30. **GASSER** **Gasser's ganglion** – the sensory ganglion of the trigeminal nerve
31. **GEROTA** **Gerota's fascia** – covers the perinephric fat capsule
32. **GOLGI** **Golgi complex** – a cellular organelle
Golgi tendon organ – a proprioceptor found in tendons
33. **GRAAF** **Graafian follicle** – the mature ovarian follicle
34. **GUYON** **Guyon's canal** – a space on the wrist through which the ulnar neurovascular bundle enters the hand
35. **HALLER** **Tripus Halleri** – the coeliac trunk, a branch of the aorta that trifurcates into three visceral branches
Haller's anastomosis – an arterial connection between the branches of the superior and inferior mesenteric artery
36. **HAVERS** **Haversian system** – the osteon, a complex of concentric lamellae of bone surrounding the Haversian canal
Haversian canal – the canal of the osteon, contains vessels, nerves and loose connective tissue
37. **HENLE** **Loop of Henle** – a loop in the nephron between the proximal and distal tubules
38. **HESELBACH** **Hesselbach's triangle** – a weakening in the anterior abdominal wall where direct abdominal hernias occur
Corona mortis of Hesselbach – an aberrant obturator artery crossing the linea terminalis and passing into the lesser pelvis
39. **HIGHMORE** **Antrum of Highmore** – the maxillary sinus, an air-filled cavity in the maxillary bone
40. **HIS** **Bundle of His** – a part of the conducting system of the heart
41. **HUNTER** **Hunter's canal** – the adductor canal, a space on the medial side of the thigh through which the femoral vessels pass
42. **HYRTL** **Hyrtl's canal** – a space containing the fibular vessels, located between the flexor hallucis longus, interosseous membrane and fibula
43. **KEITH** **Keith-Flack node** – the SA node of the conducting system of the heart
44. **KIESSELBACH** **Kiesselbach's plexus** – a vascular plexus located in the nasal septum with a high propensity to bleed
45. **KILLIAN** **Killian's triangle** – a narrowing at the transition between the pharynx and oesophagus

46. **KOHLRAUSCH** **Kohlrausch's fold** – the right transverse fold of the rectum
47. **LANGERHANS** **Islets of Langerhans** – the hormone-producing part of the pancreas
48. **LIEBERKÜHN** **Crypts of Lieberkühn** – glands of the small intestine
49. **LISFRANC** **Lisfranc's joint** – an articular line located transversely through the metatarsal bones
50. **LUSCHKA** **Foramina of Luschka** – paired lateral openings draining cerebrospinal fluid from the fourth ventricle into the subarachnoid space
51. **LUYS** **Luys' body** – the subthalamic nucleus of the diencephalon
52. **MAGENDIE** **Foramen of Magendie** – an unpaired median opening draining cerebrospinal fluid from the fourth ventricle into the subarachnoid space
53. **MALPHIGI** **Renal corpuscle of Malphigi** – the initial part of the nephron
54. **McBURNIE** **McBurney's point** – the projection of the appendix on the surface of the anterior abdominal wall
55. **MECKEL** **Meckel's cavity** – a cavity within the dura mater housing the trigeminal ganglion
56. **MECKEL** **Meckel's diverticle** – a blind variable developmental pouch located on the ventral side of the ileum
57. **MEIBOM** **Meibomian glands** – apocrine glands of the eyelid
58. **MEISSNER** **Meissner's plexus** – a nervous plexus in the submucosa of the gastrointestinal tract
Meissner's corpuscles – tactile receptors located in the dermis
59. **MONRO** **Foramina of Monro** – paired apertures connecting the lateral and third ventricles
Monro line – a line connecting the navel and right superior anterior iliac spine
60. **MÜLLER** **Müllerian duct** – the embryonic precursor of the female genital tract
61. **ODDI** **Sphincter of Oddi** – the sphincter of the bile duct
62. **PAWLIK** **Pawlik's triangle** (*area trigonalis vaginae*) – a triangular space where the anterior wall of vagina and the trigone of the urinary bladder are in contact
63. **PEYER** **Peyer's patches** – lymphoid nodules aggregated in the submucosa of the ileum
64. **POUPART** **Poupart's ligament** – the inguinal ligament
65. **PURKYNĚ** **Purkynje's (Purkyně's) fibers** – terminal branches of the conducting system of the heart located beneath the endocardium in the ventricles
Purkynje's (Purkyně's) cells – large efferent neurons within the Purkinje cell layer (*stratum purkinjense*) of the cerebellar cortex
66. **RANVIER** **Nodes of Ranvier** – gaps in the myelin sheath of neurons between Schwann's cells
67. **RATHEKE** **Rathke's pouch** – a pouch in the upper part of the stomodaeum that gives rise to the adenohypophysis
68. **RETZIUS** **Veins of Retzius** – retroperitoneal venous connections between the portal and caval venous systems
Retzius' space – a space between the pubic symphysis and the urinary bladder
69. **REXED** **Rexed laminae** – 10 layers of grey matter on the transverse section of the spinal cord
70. **ROLANDO** **Rolando's substance** – a nucleus located in the posterior horn of the spinal cord
Rolandic sulcus – the central sulcus of the cerebral cortex
71. **SANTORINI** **Santorini's papilla** – the opening of the accessory pancreatic duct into the duodenum
72. **SCARPA** **Fascia of Scarpa** – a subcutaneous fibrous layer of the anterior abdominal wall
73. **SCHLEMM** **Schlemm's canal** – a channel that drains aqueous humor into the bloodstream
74. **SCHWANN** **Schwann cells** – cells covering the axons in the peripheral nervous system
Schwann's sheath – the neurolemma, the outer layer of the Schwann cell
75. **SORGIVS** **Sorgiv node** – the sentinel lymph node of spreading breast tumours, located on the second or third slip of the serratus anterior
76. **STENSEN** **Stensen duct** – the excretory duct of the parotid gland
77. **SYLVIVS** **Aqueduct of Sylviv** – connects the third and fourth ventricles in the brainstem
Sylvian fissure – divides the frontal and parietal lobes from the temporal lobe
78. **TAWARA** **Tawara's branches** – right and left branches of the atrioventricular bundle in the conducting system of the heart
79. **TREITZ** **Muscle/ligament of Treitz** – the suspensory muscle (or ligament) of the duodenum extending from the right crus of the diaphragm to the duodenojejunal flexure
Treitz's retropancreatic membrane – the fibrous remnant of the dorsal mesoduodenum separating the pancreas from the great retroperitoneal vessels
Treitz's intersigmoid recess – a peritoneal recess between the parietal peritoneum and the sigmoid mesocolon
80. **VATER** **Ampulla of Vater** – the common opening of the pancreatic duct and the bile duct in the duodenum
Vater-Pacinian corpuscles – tactile receptors in the skin
81. **VIRCHOW** **Virchow-Robin spaces** – pouches of the subarachnoid space located around vessels under the pia mater
Virchow-Troisier lymph nodes – the left supraclavicular lymph nodes
82. **WALDEYER** **Waldeyer's ring** – a group of tonsils arranged in a ring in the pharynx
83. **WERNICKE** **Wernicke's area** – an area of the cerebral cortex located in the superior temporal gyrus (Brodmann area 22) in the dominant hemisphere
84. **WHARTON** **Wharton's duct** – the excretory duct of the submandibular gland
Wharton's jelly – connective tissue in the umbilical cord
85. **WILLIS** **Circle of Willis** – a circular arterial anastomosis located at the base of the brain

I. Definitions and history

1. Explain the difference between anatomical terminology and nomenclature. (p. 2)
2. Explain the phrase "obsolete term". (p. 2)
3. Explain what an eponym is. (p. 2)
4. Explain the difference between systematic and topographic anatomy. (p. 2)
5. State the author, year and name of the publication of the first modern textbook of anatomy. (p. 2)

II. Terms and abbreviations

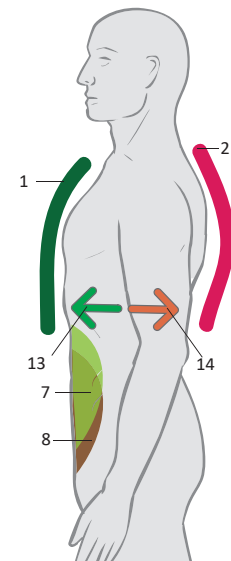
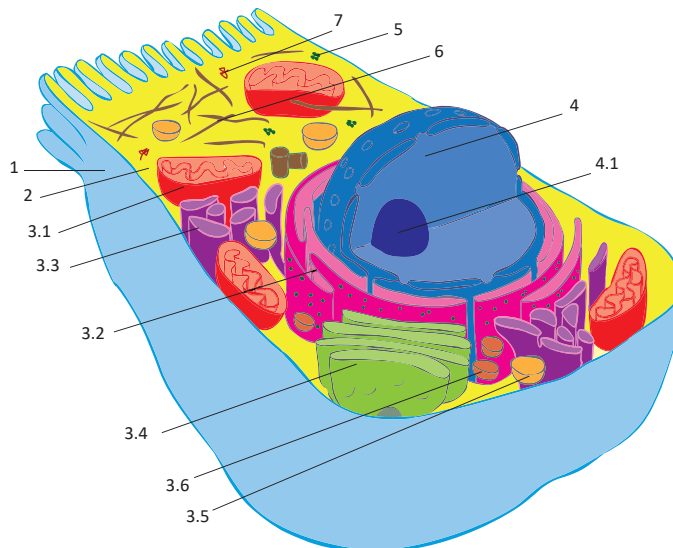
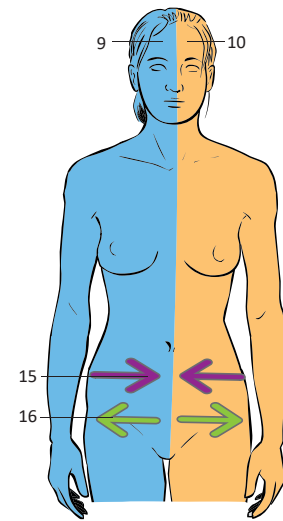
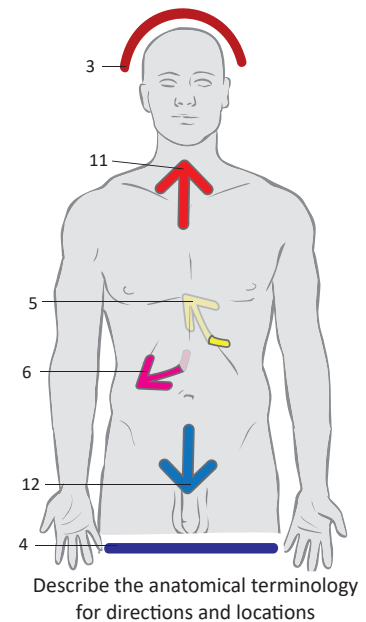
6. List all the different structures termed "sinus". (p. 3)
7. Explain the difference between the abbreviations "gl." and "gll.". (p. 3)
8. Define the abbreviations "l.sin." and "l.dx.". (p. 3)
9. Give the abbreviations of the terms "superior" and "inferior". (p. 3)

III. Histology

10. Describe the function of the smooth and granular endoplasmic reticulum. (p. 4)
11. List all the components of the cytoskeleton. (p. 4)
12. Name all the different types of epithelium and state where each type can be found. (p. 4)
13. Name three types of cartilage and describe where they can be found. (p. 4)
14. Name three types of muscle tissue and describe where they can be found. (p. 4)

IV. Embryology

15. Explain the term "primary germ layers". (p. 5)
16. Name the parts of the human body that originate from the endoderm. (p. 5)
17. List all the places to which the cells of the neural crest migrate. (p. 5)
18. State which muscles have a branchiogenic origin. (p. 5)
19. State which cranial nerves innervate which muscles. (p. 5)
20. Define the embryogenic period. (p. 5)
21. Name the three primary germ layers and some structures that originate from them. (p. 5)
22. List the muscles that originate from each pharyngeal arch. (p. 5)



V. Anatomical changes in childhood

23. Define the term "menarche". (p. 6)
24. State to which level the spinal cord extends in an adult and explain why. (p. 6)
25. State the weight of the brain of a newborn and an adult. (p. 6)
26. State all the organs that change location during development and explain why they do this. (p. 6)
27. Describe the changes of the ratio of the size of the head and trunk during development. (p. 6)

VI. Anatomical changes in puberty

28. Explain what is the Tanner scale. (p. 7)
29. Describe the shape changes of the breast in females. (p. 7)
30. List the different stages of development of the male external genital organs (p. 7)

VII. Planes and lines of the human body

31. Describe in detail the basic anatomical position. (p. 8)
32. Define the median plane. (p. 8)
33. State and define which lines go through the lateral side of the thorax. (p. 8)

VIII. Terms of direction and location

34. Define the term "contralateral". (p. 9)
35. Explain the differences in terminology of direction on the trunk, limbs and brain. (p. 9)
36. List the terms of direction and location used on the upper limb. (p. 9)
37. Define the term "aboral". (p. 9)

IX. Parts of the human body and constitutional typology

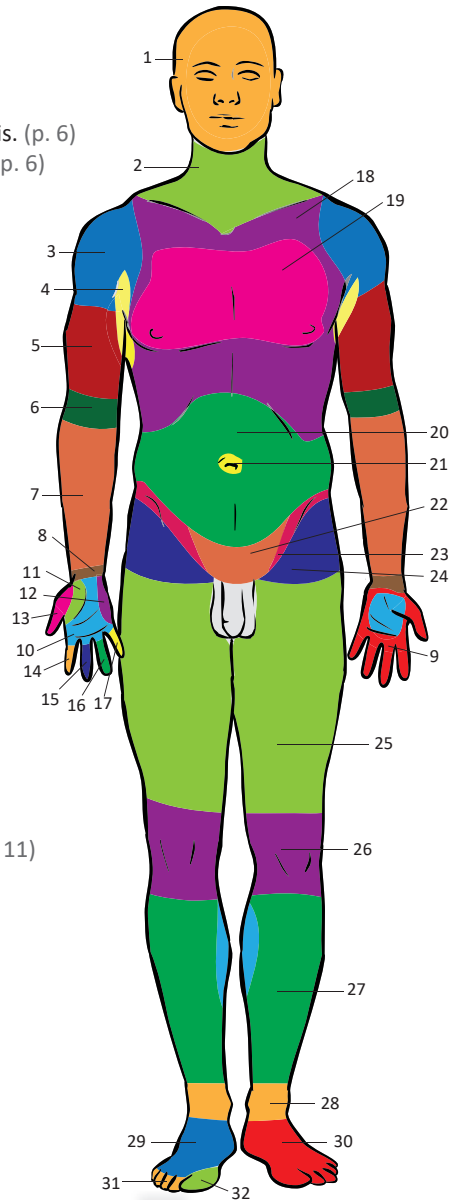
38. State the Latin term for the thumb and big toe. (p. 10)
39. Describe the asthenic body type. (p. 10)

X. Regions of the human body

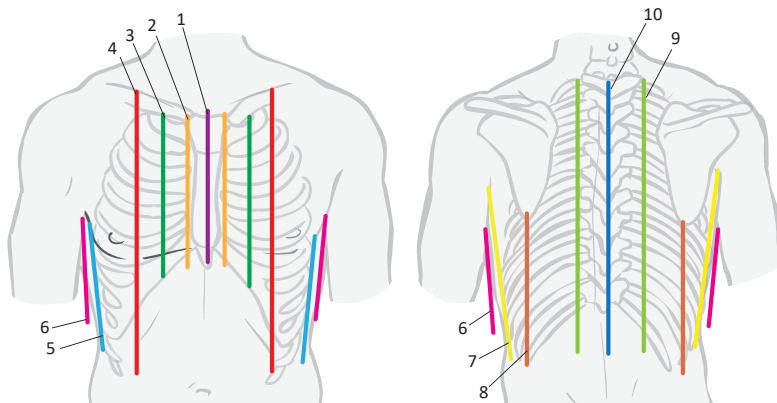
40. List the four regions of the neck and list the triangles that can be found in each region (p. 11)
41. List the six regions of the abdomen. (p. 11)

XI. Eponyms

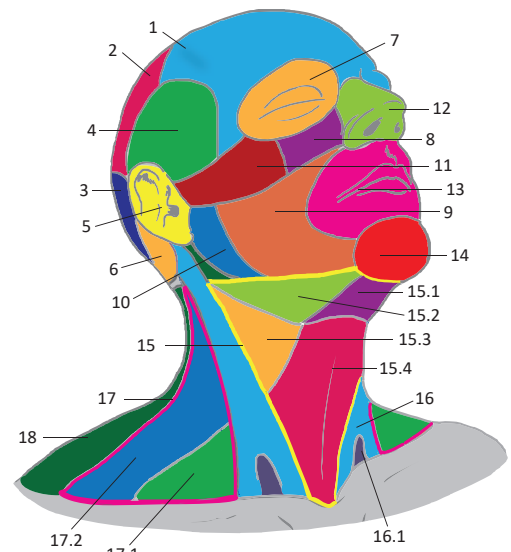
42. List two structures named after J. E. Purkyně. (p. 12)
43. List the embryonal vascular ducts and explain their function. (p. 12–13)
44. List all lymph nodes that have eponyms. (p. 12–13)
45. State all parts of the body associated with the eponym "Treitz". (p. 12)



Describe the main regions of the human body



Describe the anatomical lines of the human body



Describe the regions of the head and neck

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